HALFFULL — SYSTEM ARCHITECTURE

From user input to clinical recommendation

USER INPUT

Labs + questionnaire

CBC / ferritin / TSH CRP / glucose / creatinine electrolytes / ALAT

FDDA questionnaire

LAYER 1 — ML MODELS

11 condition classifiers

NHANES 2017-2020 training data

- Menopause
- Thyroid
- Kidney
- Anemia
- Inflammation
- Sleep disorder
- Liver
- Prediabetes
- Electrolytes
- Hepatitis B/C
- Iron deficiency

OUTPUT

P(condition) × 11

MODELS

LR, LR-L1/L2, GBM, RF

CLUSTER LAYER — NEW

Unsupervised neighbour matching

UMAP embedding + HDBSCAN

- Condition overlap
- Co-morbidity patterns
- Extended lab flags

OUTPUT

Cluster fingerprint

SIGNAL

Labs outside ref. range

BAYESIAN LAYER — NEW

Symptom questionnaire update

Triggered when P > 0.40

- Night sweats
- Mood swings
- Irregular periods
- Fatigue pattern
- Per-condition Qs

posterior odds = prior odds × LR(answer)

LAYER 2 — THRESHOLD + RANKING

Top-5 condition selection

Per-condition recall floors applied; posteriors ranked

HIGH-RISK CONDITIONS	LOWER-RISK CONDITIONS	TRIGGER THRESHOLD	OVER-ALERT CAP
Recall ≥ 85%	Recall ≥ 70%	P > 0.40 for Bayes Qs	≤ 3 FP per TP

calibrated posteriors + cluster signals

LAYER 3 — LLM SYNTHESIS

MedGemma

Receives ML posteriors + Bayesian updates + cluster fingerprint

Condition list fidelity · No hallucinated conditions · Urgency calibration · Contextual reasoning

HARD RULE	DROP TOLERANCE
No conditions added outside inputs	Must keep all P > 0.65 conditions

LAYER 4 — END-TO-END EVALUATION

Synthetic cohort validation

~600 users · 50 per condition · 50 healthy · known ground truth labels

PRIMARY METRIC	LLM DELTA	BAYES GAIN	CLUSTER NOVELTY
Top-5 coverage ≥ 90%	Coverage vs. models-only	Coverage delta post-update	Extended lab precision

USER-FACING OUTPUT

Personalised fatigue report

- Up to 5 conditions worth discussing with your doctor
- Confidence-weighted urgency framing per condition
- Extended lab tests suggested from cluster neighbours
- Symptom confirmation summary from Bayesian questions

Original architecture · Cluster layer (new) · Bayesian update (new) · LLM synthesis · Evaluation layer · Output